

Exploring the Interplay between Family Socioeconomic Status, Children’s Summer Activities, and Executive Functions

Chuu Nyan¹, Tin Q. Nguyen^{2,3}, Caroline Painter¹, Blaire M. Porter¹, and Jessica A. Church¹

¹Department of Psychology, The University of Texas at Austin, ²Vanderbilt University Institute of Imaging Science, and ³Peabody College of Education and Human Development



TEXAS
The University of Texas at Austin

Background

- Executive functions (EFs) are cognitive processes that are fundamental to goal achievement. These processes are slow to mature and can be influenced by many factors such as early experiences and environments¹.
- EF performance is associated with family socioeconomic status (SES); quality education and care enhance, but poverty and stress diminish the EF outcome¹. These associations may be more apparent during the summer when children have less consistent access to various learning opportunities.
- Prior studies have highlighted significant decline in working memory and academic skills development among children from low SES families during the summer^{2,3}.
- Children who spent time in less-structured daily activities had higher EFs⁴. However, the interaction between summer activities, SES, and EFs are not fully explored.

Research Questions

RQ1: Are there correlations between SES, summer activity, and EF?

RQ2: How do family SES & summer activity relate to post-summer EF?

Methods

Participants

- Our sample included $N = 200$, wherein $n = 128$ had data for summer activities & EF tests.
- 98 4th graders & 102 7th graders; 96 females, 96 males, 7 others, 1 prefer not to say.
- Pre-summer visit is between April-May; post-summer visit is between August-September.

Family SES (Pre-summer visit): Parental self-report.

Item	SES Measure	median	sd
1	Household Income	170,000	235,079.20
	Household Income <i>Ratio</i>	40,000	69,705.10
2	Mother's Education (in years)	19 (Some graduate or professional school but did not receive a degree)	2.09
3	Father's Education (in years)	18 (Bachelor's degree)	2.57

Summer Activity (Post-summer visit): DCN Summer Activity questionnaire.

Item	Academic Activity	<i>m</i>	<i>sd</i>	Never	One time	Every month	Weekly	Daily
1	read for enjoyment	2.56	1.31	11%	13%	13%	35%	28%
2	go to tutoring	0.36	0.95	86%	2%	3%	8%	1%
3	play educational games/video games	1.95	1.47	30%	5%	18%	33%	14%
4	participate in organized academic activities	0.54	1.14	76%	5%	3%	10%	3%
5	participate in other academic activities	1.10	1.40	57%	5%	15%	17%	6%

Item	Social Activity	<i>m</i>	<i>sd</i>	Never	One time	Every month	Weekly	Daily
1	hangout indoors with friends	2.27	0.99	9%	35%	43%	5%	
2	hangout online with friends	3.01	1.23	10%	2%	3%	36%	43%
3*	hangout indoors with siblings	3.25	1.39	14%	0%	2%	15%	69%
4	attend a religious service	0.95	1.28	61%	6%	12%	19%	2%
5	volunteer	0.76	1.01	56%	19%	18%	5%	2%
6	attend an organized group	0.82	1.20	63%	9%	13%	12%	2%

Item	Physical Activity	<i>m</i>	<i>sd</i>	Never	One time	Every month	Weekly	Daily
1	participate in outdoor activities	2.85	0.94	2%	5%	22%	45%	25%
2	participate in organized sports	1.19	1.42	53%	8%	10%	23%	5%
3	participate in recreational sports	1.59	1.47	40%	9%	14%	29%	9%
4	play outside with friends	2.36	1.01	8%	8%	34%	42%	8%
5*	play outside with siblings	2.28	1.47	22%	5%	20%	28%	25%

Item	Leisure Activity	<i>m</i>	<i>sd</i>	Never	One time	Every month	Weekly	Daily
1	practice a musical instrument	1.08	1.49	60%	8%	8%	14%	11%
2	play video games	3.18	1.18	7%	2%	13%	22%	56%
3	go to the movies	1.33	0.95	26%	22%	44%	8%	0%
4	attend a live sports game/match	2.28	0.94	69%	23%	4%	4%	0%
5	go on vacation	0.44	0.76	8%	7%	39%	42%	4%

Item	Camp Activity	<i>m</i>	<i>sd</i>	Never	A couple days	One week	Two weeks	One month	Whole Summer
1	go to an educational sleepaway camp	0.25	0.79	89%	2%	7%	1%	1%	1%
2	go to an educational day camp	0.75	1.29	71%	4%	12%	7%	5%	1%
3	go to a non-academic sleepaway camp	0.98	1.36	62%	2%	18%	13%	5%	1%
4	go to a non-academic day camp	1.88	1.65	36%	5%	18%	22%	15%	5%

Executive functions (Post-summer visit): Six tasks⁶.

- 2 working memory, 2 inhibitory control, 2 cognitive flexibility tasks.

Analysis in R: Correlation analyses were conducted to analyze the correlations between activities and SES, and activities and EF. **Regression analyses** were conducted to analyze the both activities and EF while controlling for SES, and the interaction between SES and summer activities in predicting EFs, all adjusting for grade and biological sex. **Post-hoc analyses** were conducted for each activity type to determine directions.

Results

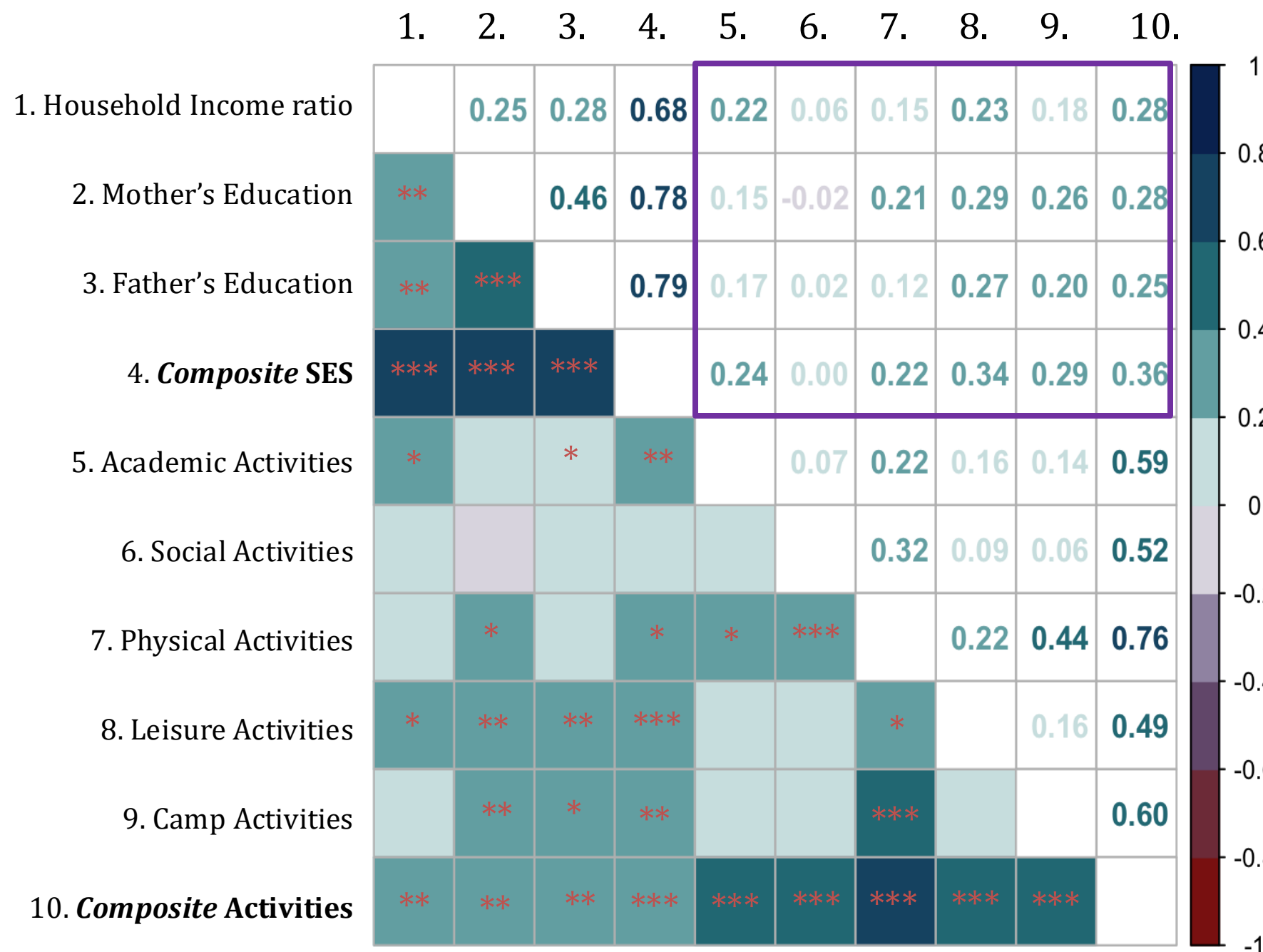
RQ1a: Family socioeconomic status was significantly correlated with multiple types of summer activities.

- ◆ Composite SES was significantly and positively correlated with academic ($p=0.007$), physical ($p=0.012$), leisure($p<0.001$), camp ($p=0.001$) and total summer activities ($p<0.001$).

- ◆ Household income ratio was correlated with academic ($p = 0.014$), leisure ($p = 0.009$), and total activities ($p = 0.002$).

- ◆ Mother's education was correlated with physical ($p = 0.017$), leisure ($p = 0.001$), camp ($p = 0.003$), and total activities ($p = 0.001$).

- ◆ Father's education was correlated with academic ($p = 0.048$), leisure ($p = 0.002$), camp ($p = 0.021$) and total activities ($p = 0.004$).



* = $p<0.05$, ** = $p<0.01$, *** = $p<0.001$

Table 2. Correlations between family SES and children's summer activities.

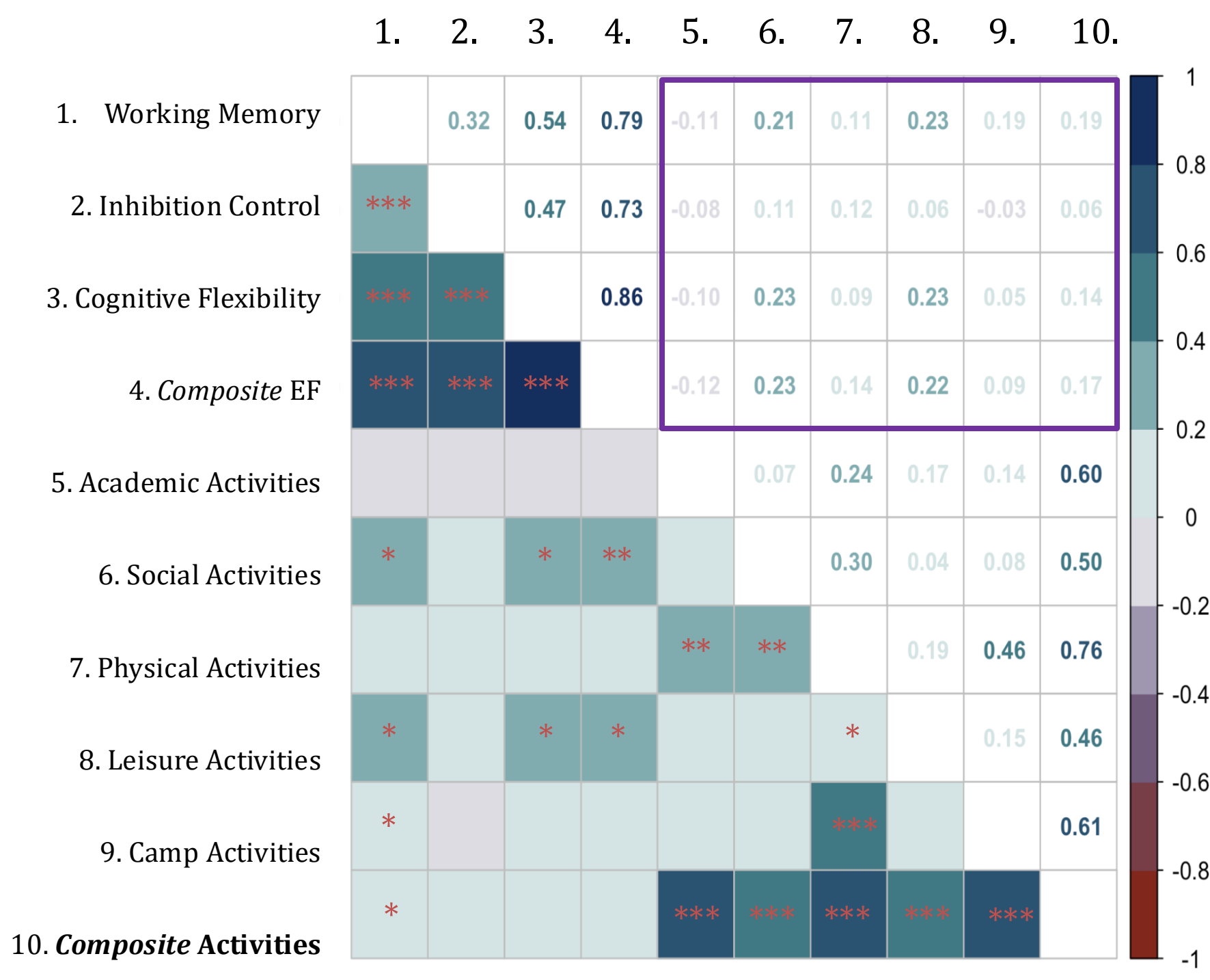
RQ1b: Social and leisure activities were significantly correlated with EF tasks.

- ◆ Social ($p = 0.008$) and leisure ($p=0.014$) activities were correlated with composite EF.

- ◆ Social ($p = 0.017$), leisure ($p = 0.011$), camp ($p = 0.037$), and total activities ($p = 0.036$) were correlated with working memory.

- ◆ Interestingly, no activities were correlated with inhibitory control.

- ◆ Social ($p = 0.01$) and leisure ($p = 0.011$) activities were correlated with cognitive flexibility.



* = $p<0.05$, ** = $p<0.01$, *** = $p<0.001$

Table 3. Correlations between children's summer activities and EFs.

RQ2a: Children's summer social and leisure activities predicted EFs, and family SES interacted with children's physical and camp activities to predict EFs.

<i>Predicting EFs</i>	Summer Activities + SES			Summer Activities x SES		
	<i>b</i>	<i>se</i>	<i>p</i>	<i>b</i>	<i>se</i>	<i>p</i>
Overall Activities	0.092	0.050	0.067	-0.091	0.048	0.060
<i>Components</i>						
Academic Activities	-0.040	0.051	0.427	-0.011	0.053	0.841
Social Activities	0.106	0.049	0.032	0.063	0.060	0.297
Physical Activities	0.089	0.047	0.061	-0.129	0.055	0.020
Leisure Activities	0.112	0.050	0.027	-0.076	0.052	0.148
Camp Activities	0.029	0.047	0.535	-0.247	0.063	0.000

Table 4. Regressions for summer activities predicting EFs controlling for SES [left], and summer activities interacting with SES to predict EFs [right].

Results

RQ2b: The interaction between family SES and summer physical or camp activities significantly predicted post-summer EFs.

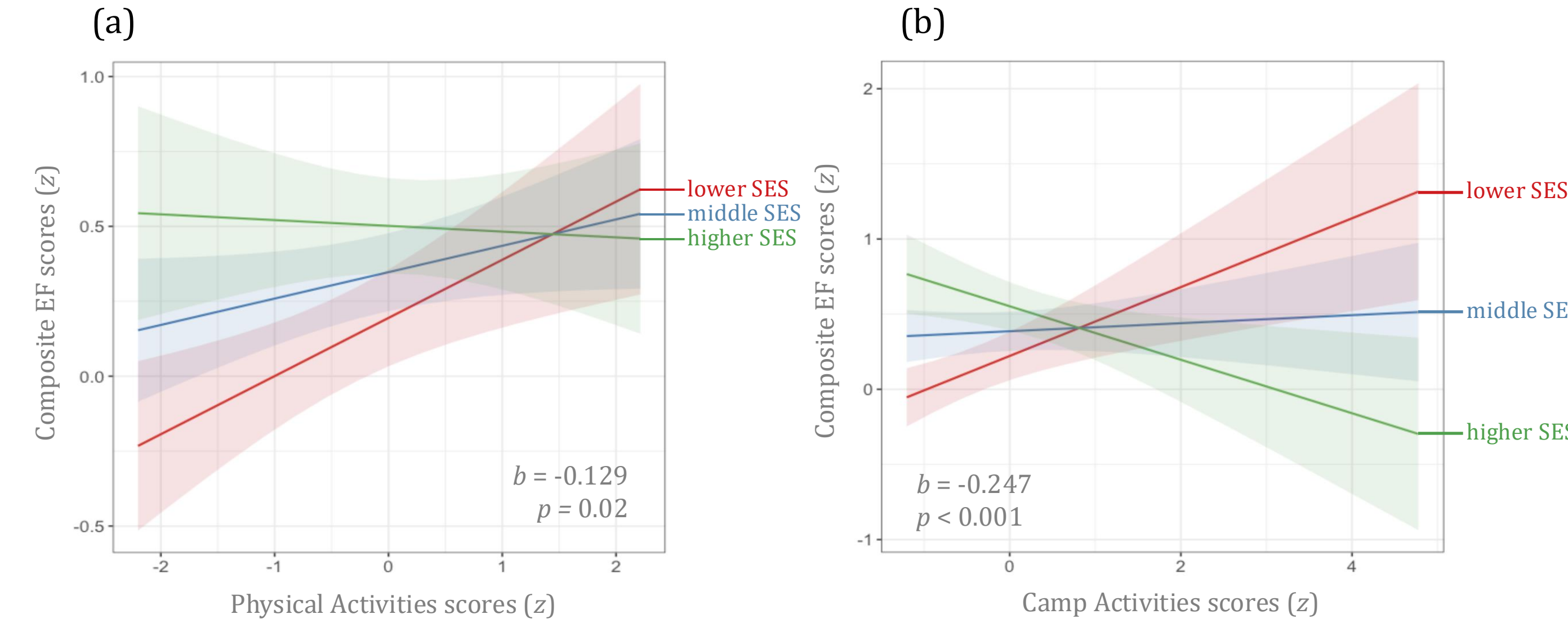


Figure. Plots of the significant interactions between (a) summer physical and (b) camp activities and SES levels for predicting EFs.

Conclusion & Implications

- ✓ In this study, we explored the interplay between family SES, summer activities, and EFs. Children's EF outcomes were associated with their summer activities, and some interacted with their family SES.
- ✓ Composite SES was strongly correlated with summer activities, and EF performance was linked to social and leisure activities during the summer. While working memory and cognitive flexibility were strongly correlated with summer activities, inhibitory control was not.
- ✓ Summer social and leisure activities were related to EFs independent of SES. This is consistent with prior findings that activities like sports positively impact EFs, even after controlling for SES⁶. Our results suggest that children who engage in more social and leisure activities tend to have stronger EFs, regardless of SES backgrounds.
- ✓ Interactions between family SES and summer physical or camp activities were significant predictors of EFs. These results align with the literature on the associations between sedentary behavior, physical activity, and EFs^{7,8}. Our study suggests that children from lower-SES backgrounds who participate in more physical and camp activities tend to have better EF outcomes.
- ✓ Current results are only preliminary as we continue to process this year's post-summer data ($n = 53$). Additionally, while our SES distribution was broad, we are limited in our collection of data from extremely low SES participants.
- ✓ **Take home:** Understanding how children's activities and family SES impact EF outcomes could offer insights for parents, educators, and policymakers to narrow the academic achievement gap among families with different SES.

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☺ ☺ ☺ I'm applying for graduate PhD programs for the 2025 cycle! ☺ ☺ ☺

For questions or more information on this project, please contact Chuu Nyan via email: chuu.nyan@utexas.edu